

1 NEIGHBORHOOD LEVEL INFLUENCE

With the current government shutdown and subsequent rippling effects, one of the most impacted aspects of low socioeconomic neighborhoods would be the limited access to affordable healthy foods. In many of these neighborhoods, residents are severely limited to unhealthy fast foods for their daily nutrition. This condition has always been an issue in which residents have limited access to grocery stores, markets, or fresh produce based on the area in which they live. Access to affordable, nutritious foods have always been a core factor in determining the health and well being of the community. This structural inequality in the food environment stems from barriers shaped by socioeconomic and spatial inequalities. Research has consistently proven that neighborhood food environments directly affect diet quality and food related illnesses. Residents in these low income areas often face multiple obstacles in healthy eating, such as economic constraints, limited transportation, and unsafe walking conditions. This all reduces the people's ability to access healthy foods even when healthy food options are available (Walker et al., 2010). All of these conditions slowly but surely will lead to food related illnesses such as diabetes or cardiovascular disease. With such limited access to affordable healthy foods, people will rely on junk food and fast food, which is a leading cause of obesity (Beaulac, Kristjansson, & Cummins, 2009). Based on the reading we know that addressing such environmental determinants needs the inclusion of the community, such as the "Community Mosaic" application. This application shows that the community together can advocate for structural change, and digital tools can assist in this development (Parker et al., 2012). We can take a look at low socioeconomic communities and apply our understanding of this to the lack of healthy food options to begin formulating possible ways to alleviate the problem, such as developing technologies to help pool the community together to advocate for better food options, creating a way to spread better nutritional education to the masses, or develop a guide in which people will be able to find healthy foods on their own. Limited access to affordable healthy foods is a powerful neighborhood level barrier that affects many

communities and severely hinders the well being of the people living in these communities. Addressing this issue requires solutions that focus on restoring the communities' ability to support their own healthy environments.

2 DESIGN CONCEPTS

2.1 Concept 1

- Idea: Create a community food access and sharing system within Nextdoor where residents can locate, share, and coordinate access to healthy food and resources.
- Features:
 - Interactive community map marking grocery stores, food pantries, farmer's markets, and mutual aid food exchange
 - "Request" and "Offer" functionalities for food sharing
 - Group buying coordination
- This design is focused on improving physical health and community social interactions by strengthening community engagement in healthy food procurement through group coordination. It encourages community participation and trust building in low SES areas where trust may be low.
- This builds on Parker et al's Community Mosaic idea of community generated health knowledge and visual mapping and addresses structural access barriers such as transportation and affordability by Walker et al (2012)(2010).

2.2 Concept 2

- Idea: Create a community improvement and feedback system within Nextdoor that promotes better healthy foods and influences local food stores for healthier food availability
- Features:
 - Health Score Ratings for stores and restaurants based on community ratings and nutritional data
 - Petition and feedback tools to advocate for affordable and healthier foods
 - Community feedback dashboard of retailers
 - Guidance tool on improving store health conditions

- This design is focused on improving environmental conditions and civic wellbeing through improved neighborhood food locations based on health advocacies. It is meant to push retailers to improve conditions of food and offerings to improve consumer options. In addition it also creates a community that engages in civic control over health.
- This uses Parker et Al's emphasis on community empowerment and structural changes of the environment (2012).

2.3 Concept 3

- Idea: Create a community food recipe swapping forum within Nextdoor that allows users to swap recipes for better food access and nutritional guidance.
- Features:
 - Healthy/Cost Effective Recipe Catalogue of community uploaded meals with flexibility on cost, ingredients, and dietary needs based on filters
 - Local ingredient finder based on the ingredients in recipe with a locator based on nearby affordable stores
 - Meal Sharing feature that allows neighbors to share meals or plan events to meal prep together
- This design focuses on improving nutritional well being and social well being across the community and neighborhood. The recipe catalog revolves around spreading healthier affordable eating habits using locally available foods. It is meant to encourage learning healthy cooking within the community.
- Addresses nutrition education gaps stated by Beaulac et al by offering healthy eating education within residents' economic and cultural context (2009).

3 CHOSEN CONCEPT

My chosen concept for the Nextdoor redesign would be the integration of the community food recipe swapping forum. This idea reimagines the app as a space in which residents of low socioeconomic neighborhoods can share, learn, and trade recipes using locally accessible healthy foods and where they can also come together to cook and share meals. In addition, the app will also offer a local ingredient finder that can help individuals find better local sources of ingredients. This design focuses on addressing the neighborhood level barrier of

limited access to affordable healthy foods, by expanding nutritional knowledge and encouraging community based learning and sharing of foods. The recipe sharing, local ingredient finder, and collaborative cooking will elevate Nextdoor to combat the issue that low SES communities have with finding healthy foods.

3.1 Feature 1 Recipe Catalogue and Upload Hub

This feature will be a dedicated recipe catalogue added to Nextdoor's navigation bar where users can browse recipe cards with images, brief descriptions, and tags for cost levels, dietary restrictions, and cultural cuisine type. Each recipe will be linked to a detailed page showing the ingredient list, preparation steps, ratings, and cost. Please look at Appendix 6.2 for images of the recipe catalogue and recipe page. The feature aims to help residents create a locally shared system in which food education is spread by sharing affordable healthy recipes while advocating community involvement. This feature further aligns with the argument that "everyday knowledge generated by community members can improve local health ecosystems" (Parker et al., 2012).

3.2 Local Ingredient Finder

The local ingredient finder is designed to allow the user to find ingredients from local sources, such as grocery stores, farmers' markets, or other low income food options easier. The locations of these places would be mapped in Nextdoor and the user can filter through options of specific ingredients based on distance, price, eligibility for SNAP/WIC benefits as needed. Please look at Appendix 6.3 for images of the local ingredient finder. The feature is designed to help mitigate spatial and economic barriers against healthy eating. The availability of fresh healthy foods is the main issue that many low SES communities are dealing with, and this feature connects the recipe catalogue to a function that helps assist users with real world availability locating. It follows the belief that the design is meant to bridge the gap between information and infrastructure and acknowledge that knowledge alone is insufficient without accessible resources (Robert Wood Johnson Foundation, 2017).

3.3 Event Feed

The last feature would be a social component that allows for neighbors to collaborate together for events or meal sharing. This aspect would be an accent that does not necessarily

involve the recipe catalog but is designed for residents to create a stronger community. Residents can host low cost meal prep gatherings, cooking demonstrations, nutrition education sessions, or work out meal sharing plans. The feature would be more of a post forum similar to the current Nextdoor post feed, but would be meant for food related events. Please look at Appendix 6.4 for images of the event feed page. This feature works to enhance social involvement within the community, while lessening the strain on individual households. It works toward reducing isolation and providing low pressure peer support along with healthy cooking to improve neighborhood well being. The cooking community events function as health interventions across the community by leveraging digital technologies to advocate for better healthy living.

4 NON TECHNOLOGICAL INNOVATION

At the meso level, to strengthen the community recipe swapping forum, a community health advocacy program can help bridge the digital and in person engagement. This program would be designed to recruit and train individuals within the community to act as experts or ambassadors who coordinate and host events to increase awareness of the application as well as spreading more nutritional knowledge. They can also help with hosting cooking demonstrations, finding new recipes to post, and help people who have difficulties with technology to learn how to use the application. By supplementing the use of technology with human interactions, we can help break down the barriers that may cause people to turn away from the app. This directly correlates to Parker et al who explains that sustainable community health design requires strong social infrastructures that support community engagement (2012). The health advocacy program provides more visibility and shared learning opportunities to the community while encouraging more engagement from individuals within the neighborhood. The recipe forum together with actual human involvement comes together to push not only digital awareness and healthy cooking to becoming an active community relying on one another for healthier foods and overall better community well being.

5 REFERENCES

1. Beaulac, J., Kristjansson, E., & Cummins, S. (2009). *A systematic review of food deserts, 1966–2007. Preventing Chronic Disease*, 6(3).
2. Parker, A. G., Harper, R., Sellen, A., & others. (2012). *Community Mosaic: Designing for health information sharing in the community*.
3. Walker, R. E., Keane, C. R., & Burke, J. G. (2010). *Disparities and access to healthy food in the United States: A review of food deserts literature. Health & Place*, 16(5), 876–884.
4. Robert Wood Johnson Foundation. (2017). *Building a Culture of Health: A Policy Framework for Action*.

6 APPENDIX

6.1 Prototype Link (Figma Sign In Required)

<https://www.figma.com/proto/o5Gb6MAX6cIb3mf0VmQHEN/NextDoor-Redesign?node-id=1-2&t=wo12VU4W7S5865Ed-1>

6.2 Images of Feature 1

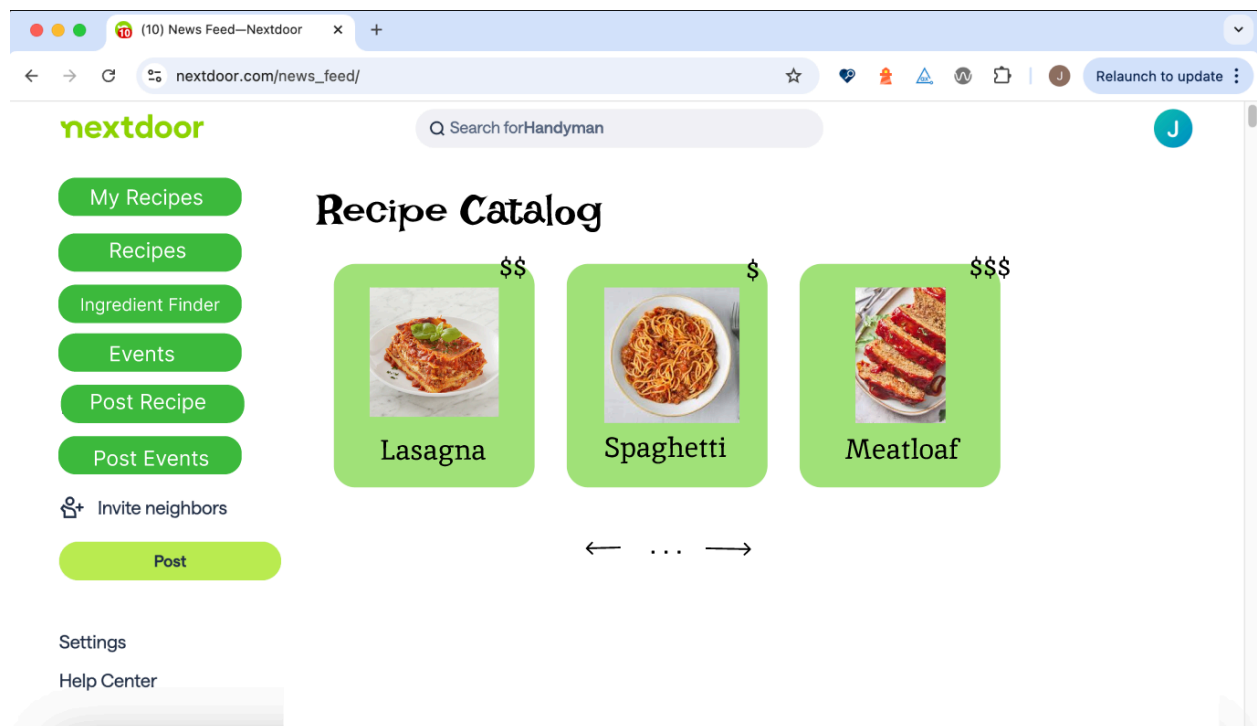


Figure 1: Recipe Catalog

6.4 Images of Feature 3

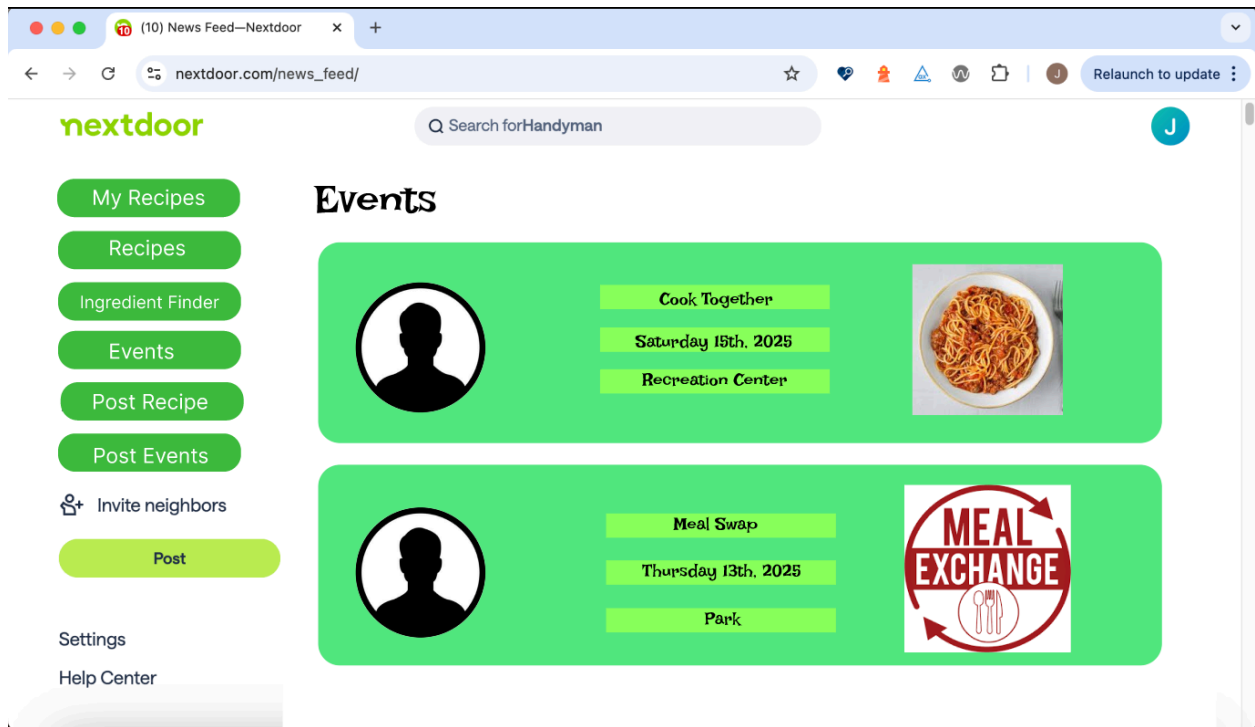


Figure 4: Event Feed Page

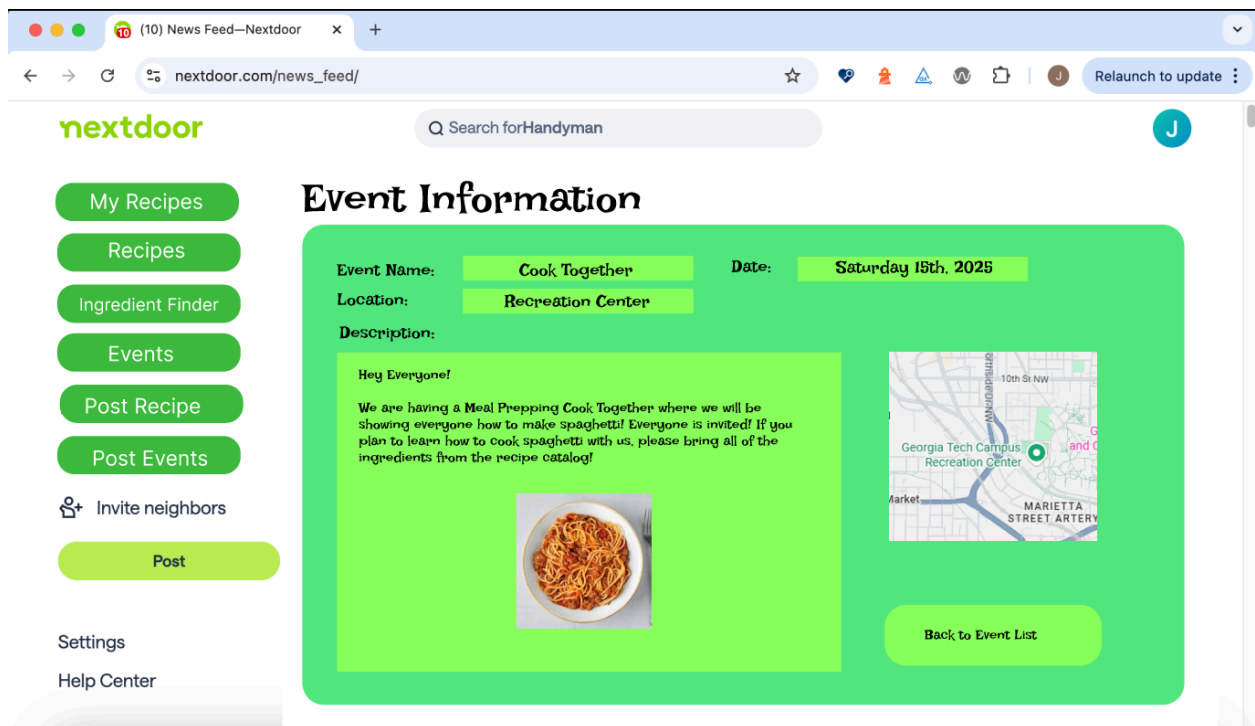


Figure 5: Event Information Page

I have not used generative AI to prepare this assignment.